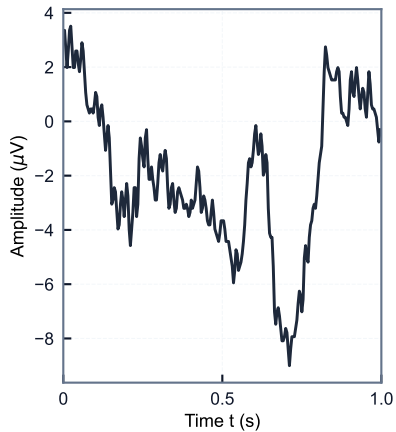
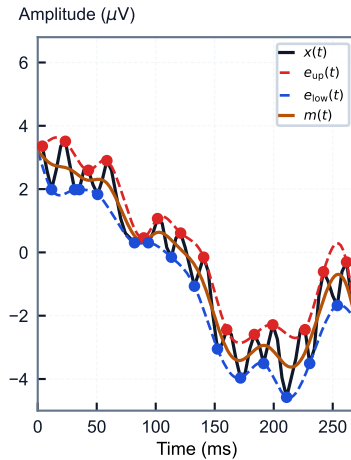


(a) Raw Non-Stationary EEG $x(t)$



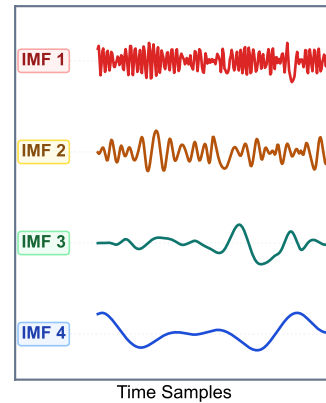
Extrema

(b) Cubic Spline Sifting Loop



Sift Modes

(c) Intrinsic Modes (IMF 1–4)



Compute

(d) 6 Non-Linear Descriptors

1. Energy: $E_k = \sum |IMF_k|^2$
2. Std Dev: $\sigma_k = \text{std}(IMF_k)$
3. Skewness: $S_k = \mathbb{E}[(IMF - \mu)^3] / \sigma^3$
4. Kurtosis: $K_k = \mathbb{E}[(IMF - \mu)^4] / \sigma^4$
5. Wiener Ent: $H_k = \frac{\exp(\mathbb{E}[\ln |IMF|^2])}{\mathbb{E}[|IMF|^2]}$
6. PSD RMS: $RMS_k = \sqrt{\frac{1}{M} \sum P_k(f)}$

Concatenate

(e) EMD Tensor $\mathbf{x}_{EMD}$

1. Mode Vector:
 $\mathbf{e}_k = [E_k, \sigma_k, S_k, K_k, H_k, RMS_k]^\top$
2. Channel Vector:
 $\mathbf{e}_c = [\mathbf{e}_1^\top, \mathbf{e}_2^\top, \mathbf{e}_3^\top, \mathbf{e}_4^\top]^\top$
3. Channel Dim:
 $\dim(\mathbf{e}_c) = 4 \times 6 = 24D$
4. Multichannel:
 $\mathbf{x}_{EMD} = [\mathbf{e}_1^\top, \dots, \mathbf{e}_C^\top]^\top$

$$\mathbf{x}_{EMD} \in \mathbb{R}^{C \times 24}$$

($C = 18 \rightarrow 432D$)